

BatteryMFormer: Multi-level Learning for Battery Degradation Trajectory Forecasting

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Abstract

Early battery degradation trajectory forecasting (BDTF), which predicts the full-life state-of-health trajectory from early operational data, is critical for battery optimization, manufacturing, and deployment. Battery degradation data exhibit two key characteristics. First, degradation data present a multi-level structure, including regularities shared within aging conditions and trajectory patterns shared across batteries. Second, degradation-related variations in voltage-current profiles are often localized to specific state of charge (SOC) intervals. Existing approaches often fail to explicitly model these characteristics. To bridge this gap, we propose BatteryMFormer, a multi-level Transformer for early BDTF. BatteryMFormer integrates (1) an aging-condition-aware decoder that injects aging-condition priors via aging-condition-informed queries and aging-condition-aware attention, (2) a meta degradation pattern memory that learns and retrieves trajectory prototypes to guide long-horizon forecasting, and (3) a dual-view encoder that jointly captures temporal dynamics and SOC-localized variations from voltage and current time series. Extensive experiments on four battery domains show that BatteryMFormer consistently outperforms state-of-the-art baselines, marking a significant step toward reliable BDTF. Our code is available at <https://github.com/Ruifeng-Tan/BatteryMFormer>.

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CCS Concepts

• Information systems → Data mining.

Keywords

materials informatics, battery informatics, time series

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1 Introduction

Rechargeable batteries are ubiquitous in modern industry, powering applications ranging from electric vehicles and grid-scale energy storage to portable electronics [13, 35, 40, 53]. In 2024, global battery shipments exceeded 1545 GWh and are projected to reach 4700 GWh by 2030 [10, 55]. This rapid expansion highlights the need for advanced modeling frameworks to support battery optimization, manufacturing, and deployment [3, 20, 33, 37, 51]. In particular, battery degradation trajectory forecasting (BDTF), which predicts battery state-of-health (SOH) trajectories from beginning of life to end of life, occupies a critical frontier. By forecasting full-life degradation trajectory from early-stage operational data, BDTF enables accelerated degradation assessment and timely maintenance for battery-powered systems [14, 22, 37].

Machine learning (ML) models have recently emerged as promising solutions to BDTF. Existing approaches primarily fall into feature-engineering-based methods and representation-learning-based methods. Feature-engineering-based methods typically extract handcrafted descriptors from voltage and current time series



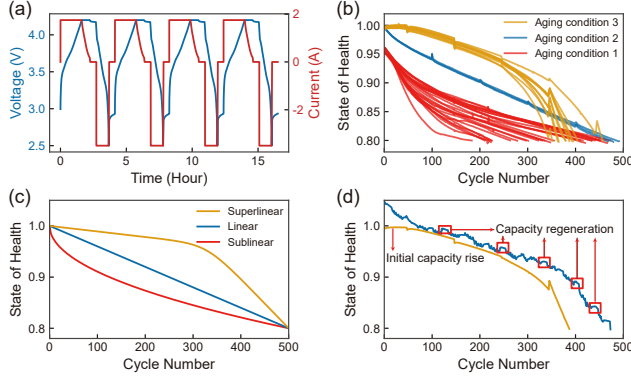


Figure 1: Motivation for multi-level learning in BDTF. (a) An example of partial operational voltage and current time series. (b) SOH trajectories under different aging conditions. (c) Schematic of three canonical trajectory shapes. (d) Examples of additional trajectory phenomena beyond the canonical shapes.

(Figure 1a) [19, 28, 41], whereas these features are often protocol-specific or dataset-specific and may be unavailable or ineffective across diverse aging conditions. Representation-learning-based methods instead focus on learning mappings from raw measurements to future SOH trajectories [14, 15, 22, 23, 25, 34, 36, 37]. An intuitive modeling choice is to treat BDTF as generic time-series forecasting and extrapolate future SOH from historical SOH using generic time series forecasters (e.g. Informer [56]) [22, 23, 25, 34, 37]. While effective in some settings, early-cycle SOH can be nearly indistinguishable across batteries whose long-horizon trajectories diverge substantially, and therefore forecasting with SOH as the only input can be unsuitable for early BDTF (Figure 1b). This limitation has motivated growing interest in models that exploit fine-grained voltage–current profiles for forecasting [14, 15, 36].

Despite these advances, current models still exhibit two critical research gaps. First, these methods operate at the *battery level* and do not explicitly model the *multi-level structure* of degradation. Batteries under the same aging condition (e.g., specifications, formation, and operating conditions) exhibit consistent operational patterns, and prior work shows that batteries under similar aging conditions can be characterized by a small set of handcrafted descriptors [18, 23, 33, 41, 45]. However, existing models fail to promote *aging-condition-consistent* representations. Moreover, although trajectories appear diverse, established battery knowledge [2] suggests that their *global shapes are highly structured* and often fall into a small family of patterns linked to common mechanisms (Figure 1c). Additional phenomena such as initial capacity rise [33] and capacity regeneration [15] can occur (Figure 1d), but the space of plausible trajectories remains constrained. Second, degradation-relevant variations in voltage–current profiles often concentrate within specific SOC intervals (Figure 2), as underlying electrochemical mechanisms can be manifested as localized electrochemical signal variations along the SOC axis (e.g., phase transition) [5, 37]. Nevertheless, most methods either emphasize temporal modeling or treat SOC intervals uniformly, diluting localized signals.

To address these limitations, we propose **BatteryMFormer** (Battery Multi-level Transformer), a novel deep learning architecture that integrates multi-level learning across aging conditions, trajectory patterns, and battery-specific representations. BatteryMFormer consists of three major components: (1) Aging-condition-aware decoder that injects aging-condition priors via aging-condition-informed queries and aging-condition-aware attention to promote aging-condition-consistent representations; (2) Meta degradation pattern memory that learns and retrieves prototypical trajectory patterns to guide long-horizon forecasting; and (3) Dual-view encoder that captures complementary temporal dynamics and SOC-localized variations from voltage–current profiles.

The main contributions of this paper are summarized as follows:

- We identify and formalize the multi-level structure of early BDTF, including aging-condition regularities, trajectory patterns shared across batteries, and SOC-localized degradation signatures in operational data.
- We propose BatteryMFormer, a multi-level Transformer that integrates (i) an aging-condition-aware decoder, (ii) a meta degradation pattern memory, and (iii) a dual-view encoder with temporal and SOC perspectives.
- We conduct extensive experimental evaluation, the results from which demonstrate the superior performance of our approach across four battery domains from the largest public real-world battery lifetime database.

2 Preliminaries

2.1 Aging Condition

We use aging condition to denote the recorded experimental settings and battery specifications that determine a battery’s degradation regime. In this work, an aging condition is represented as a tuple of aging factors, including positive electrode, negative electrode, electrolyte, package structure, nominal capacity, manufacturer, formation protocol, charge protocol, discharge protocol, and operating temperature. Different factor tuples correspond to different aging conditions. Batteries operated under different aging conditions can exhibit distinct degradation trajectories (Figure 1b) and patterns of voltage–current profiles (Figure 2) [35, 36, 51].

2.2 Degradation Trajectory

Degradation trajectories are measured from repeated cycles, with each having a charge and discharge process. Following prior work [27, 33, 36], we compute the discharge capacity of cycle i as

$$Cap_i = \int_{t_1}^{t_2} |I(t)| dt, \quad (1)$$

where t_1 and t_2 denote the start and end times of the discharge process, and $I(t)$ is the measured current at time t , with $|I(t)|$ used to make the definition invariant to sign conventions. The state of health (SOH) at cycle i is defined as

$$SOH_i = \frac{Cap_i}{Cap_0 \times DoD}, \quad (2)$$

where DoD is the depth of discharge, and Cap_0 denotes the nominal capacity for all datasets except CALB, where Cap_0 is defined as

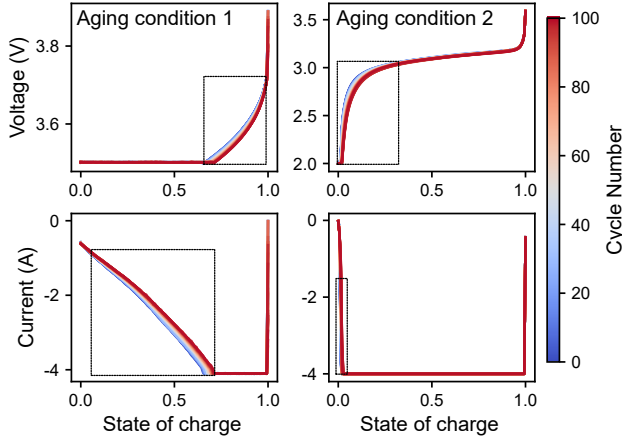


Figure 2: SOC-localized degradation signatures in voltage-current profiles. Voltage–SOC (top) and current–SOC (bottom) curves over the first 100 cycles for two representative aging conditions. Although the global profiles evolve smoothly with cycling, aging-induced deviations can concentrate within specific SOC intervals (dashed boxes).

the first-cycle discharge capacity following the CALB protocol in BatteryLife [36].

2.3 Task Formulation

Following prior work [33, 36, 51], we use the first $S \leq 100$ cycles as the early stage and forecast the SOH trajectory beyond the observation window. We denote by $\mathbf{a}$ the available aging-condition metadata of a battery, including recorded experimental settings and specifications. Let $\mathbf{X}_i$ denote the cycle- i operational data, consisting of voltage and current time series (and any auxiliary variables derived from $\mathbf{a}$ and these early-cycle measurements, e.g., capacity and SOC). We define the early input as ordered sequences

$$\mathbf{G}_{1:S} = (\mathbf{X}_{1:S}, \mathbf{a}), \quad \mathbf{X}_{1:S} = [\mathbf{X}_1, \dots, \mathbf{X}_S]. \quad (3)$$

We use t_{eol} to denote the end-of-life (EOL) cycle index, defined as the first cycle at which SOH falls below a threshold τ (Appendix A). Let $\mathbf{y}_{1:t_{\text{eol}}} \in \mathbb{R}^{t_{\text{eol}}}$ denote the measured SOH trajectory. The goal of early BDTF is to learn a forecasting model $f(\cdot)$ that predicts the future SOH trajectory given the first S cycles:

$$\hat{\mathbf{y}}_{S+1:t_{\text{eol}}} = f(\mathbf{G}_{1:S}). \quad (4)$$

3 Methodology

Figure 3 presents the overall architecture of BatteryMFormer, a Transformer with multi-level inductive biases for early BDTF. BatteryMFormer encodes early operational data into complementary temporal and SOC tokens via a dual-view encoder (Section 3.1), refines these tokens with an aging-condition-aware decoder (Section 3.2), and retrieves prototypical trajectory patterns from a meta degradation pattern memory (Section 3.3) to guide long-horizon forecasting.

3.1 Dual-View Encoder

The dual-view encoder maps early operational data into temporal-view and SOC-view tokens. Following BatteryLife [36], we obtain within-cycle capacity via ampere-hour counting from current time series to encode temporal information and additionally compute SOC (Appendix B). After resampling each cycle to L data points, the first S cycles are represented as $\mathbf{X} \in \mathbb{R}^{S \times L \times 4}$ with variables (voltage, current, capacity, SOC).

SOC view. To capture SOC-localized degradation signatures (Figure 2), we construct SOC-view tokens by modeling cross-cycle evolution within each SOC interval. Given $\mathbf{X} \in \mathbb{R}^{S \times L \times 4}$, where each cycle contains L SOC-aligned points and 4 variables, we reshape the i -th cycle as $\mathbf{X}_i \in \mathbb{R}^{4 \times L}$, treating variables as channels. We then apply a 1D convolution along the SOC axis:

$$\hat{\mathbf{Z}}_i = \text{Conv1D}(\mathbf{X}_i) \in \mathbb{R}^{d \times M}, \quad M = \left\lfloor \frac{L - P}{P} \right\rfloor + 1, \quad (5)$$

where P is both the patch length and stride. Stacking all cycles yields $\hat{\mathbf{Z}} \in \mathbb{R}^{S \times d \times M}$. For each SOC interval m , we collect $\hat{\mathbf{Z}}_{:,m} \in \mathbb{R}^{S \times d}$ across cycles and feed it into a shared temporal encoder implemented with feed-forward neural networks and GELU activations [12]. The encoder aggregates information along the cycle axis and produces one SOC token:

$$\mathbf{t}_m^{\text{soc}} = \text{TempEnc}(\hat{\mathbf{Z}}_{:,m}) \in \mathbb{R}^d. \quad (6)$$

Concatenating all interval tokens yields $\mathbf{T}^{\text{soc}} = [\mathbf{t}_1^{\text{soc}}; \dots; \mathbf{t}_M^{\text{soc}}] \in \mathbb{R}^{M \times d}$.

Temporal view. In parallel to the SOC view, we construct a temporal view that summarizes each early cycle as a cycle-level token to capture intra-cycle dynamics. Following CyclePatch [36], we project the resampled multivariate series of cycle i into a d -dimensional embedding and refine it with an intra-cycle encoder:

$$\hat{\mathbf{X}}_i = \text{CyclePatch}(\mathbf{X}_i) = \text{Flatten}(\mathbf{X}_i)\mathbf{W} + \mathbf{b}, \quad (7)$$

$$\mathbf{z}_i^{\text{temporal}} = \text{Intra-CycleEncoder}(\hat{\mathbf{X}}_i), \quad i = 1, \dots, S, \quad (8)$$

where $\mathbf{W}$ and $\mathbf{b}$ are learnable parameters. Stacking $\{\mathbf{z}_i^{\text{temporal}}\}_{i=1}^S$ yields temporal tokens

$$\mathbf{H}^{\text{temporal}} = [\mathbf{z}_1^{\text{temporal}}; \dots; \mathbf{z}_S^{\text{temporal}}] \in \mathbb{R}^{S \times d}. \quad (9)$$

We further inject cycle-level descriptors by projecting $\mathbf{X}_f \in \mathbb{R}^{S \times d_f}$ to the token space and adding it to $\mathbf{H}^{\text{temporal}}$:

$$\mathbf{T}^{\text{temporal}} = \mathbf{H}^{\text{temporal}} + \mathbf{X}_f \mathbf{W}_f + \mathbf{b}_f, \quad (10)$$

where $\mathbf{W}_f$ and $\mathbf{b}_f$ are learnable parameters. In this work, $\mathbf{X}_f$ consists of Coulombic efficiency and energy efficiency, which are commonly available on a per-cycle basis.

Together, $\mathbf{T}^{\text{temporal}} \in \mathbb{R}^{S \times d}$ and $\mathbf{T}^{\text{soc}} \in \mathbb{R}^{M \times d}$ provide complementary inputs for subsequent decoding.

3.2 Aging-Condition-Aware Decoder

Batteries operated under the same/similar aging conditions often exhibit consistent/similar degradation signatures [18, 33, 41, 45]. To exploit such aging-condition-level regularities, we design an aging-condition-aware decoder (ACDecoder) with two mechanisms: (i) *aging-condition-informed queries*, which inject an aging-condition

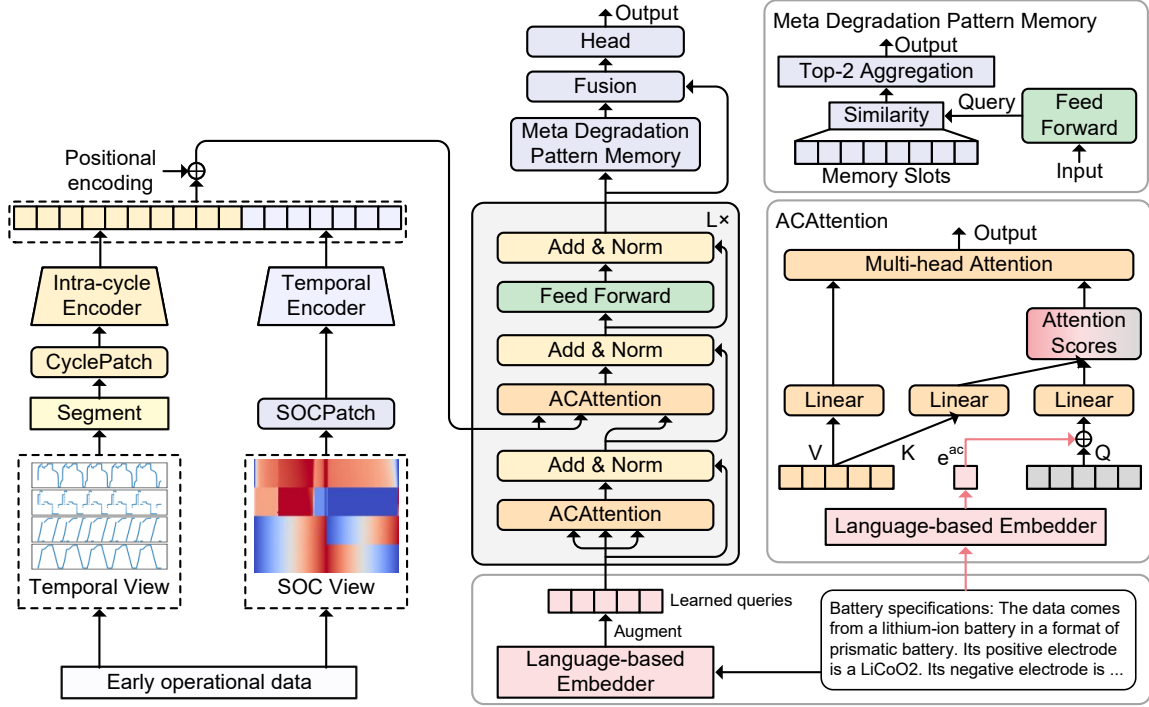


Figure 3: An overview of BatteryMFormer. Left: dual-view encoder (temporal and SOC views). Middle: aging-condition-aware decoder. Right: details of meta degradation pattern memory and aging-condition-aware attention.

prior into the decoder states, and (ii) *aging-condition-aware attention*, which conditions attention on the aging-condition prior.

Aging-condition-informed queries. Inspired by [48], ACDecoder starts from learnable generic queries $\mathbf{Q}_g \in \mathbb{R}^{s \times d}$ and injects aging-condition information via additive conditioning. Let $\mathbf{a}$ denote the structured aging-condition metadata (Section 2.1) and let $\pi(\mathbf{a})$ be the corresponding metadata-to-text prompt [35]. We encode $\pi(\mathbf{a})$ using a language-based embedder:

$$\mathbf{z}^{ac} = \text{LastValid}(\text{Enc}(\pi(\mathbf{a}))) \in \mathbb{R}^{d_{enc}}, \quad (11)$$

$$\mathbf{e}^{ac} = \mathbf{z}^{ac} \mathbf{W}_1 + \mathbf{b}_1 \in \mathbb{R}^d, \quad (12)$$

where $\text{Enc}(\cdot)$ is a language-based embedder (Qwen3-Embedding-0.6B [54]), $\text{LastValid}(\cdot)$ retrieves the embedding of the last non-padding token, and $\mathbf{W}_1 \in \mathbb{R}^{d_{enc} \times d}$ and $\mathbf{b}_1 \in \mathbb{R}^d$ are learnable parameters. We then project $\mathbf{e}^{ac}$ to produce one prior vector per query token:

$$\hat{\mathbf{e}}_i^{ac} = \mathbf{e}^{ac} \mathbf{W}_{2,i} + \mathbf{b}_{2,i} \in \mathbb{R}^d, \quad i = 1, \dots, \bar{s}, \quad (13)$$

$$\hat{\mathbf{E}}^{ac} = [\hat{\mathbf{e}}_1^{ac}; \dots; \hat{\mathbf{e}}_{\bar{s}}^{ac}] \in \mathbb{R}^{\bar{s} \times d}, \quad (14)$$

$$\mathbf{X}_{de} = \mathbf{Q}_g + \hat{\mathbf{E}}^{ac}. \quad (15)$$

Here $\mathbf{W}_{2,i} \in \mathbb{R}^{d \times d}$ and $\mathbf{b}_{2,i} \in \mathbb{R}^d$ are learnable parameters. Each $\hat{\mathbf{e}}_i^{ac}$ provides a query-specific prior, yielding aging-condition-informed queries (ACQuery) $\mathbf{X}_{de}$ for conditioning different queries on different aspects of the aging-condition information.

Aging-condition-aware attention. Beyond query initialization, ACDecoder promotes aging-condition-consistent attention by modulating queries with $\hat{\mathbf{E}}^{ac}$. Given queries $\mathbf{Q}$ and key-value tokens

$(\mathbf{K}, \mathbf{V})$, we define aging-condition-aware attention (ACAttention) as follows:

$$\text{ACAttention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}, \hat{\mathbf{E}}^{ac}) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) \mathbf{W}^O, \quad (16)$$

$$\text{head}_i = \text{Attention}((\mathbf{Q} + \hat{\mathbf{E}}^{ac}) \mathbf{W}_i^Q, \mathbf{K} \mathbf{W}_i^K, \mathbf{V} \mathbf{W}_i^V). \quad (17)$$

Here $\text{Attention}(\cdot)$ is the attention in standard Transformer [42]. This query modulation injects aging-condition priors into every attention operation, thereby promoting aging-condition-consistent decoding throughout the network.

ACDecoder layer. Let $\mathbf{T}^{\text{temporal}} \in \mathbb{R}^{S \times d}$ and $\mathbf{T}^{\text{soc}} \in \mathbb{R}^{M \times d}$ be the dual-view tokens (Section 3.1), and $\mathbf{T} = [\mathbf{T}^{\text{temporal}}; \mathbf{T}^{\text{soc}}] + \mathbf{P} \in \mathbb{R}^{(S+M) \times d}$, where $\mathbf{P} \in \mathbb{R}^{(S+M) \times d}$ is positional encoding [42]. With $\mathbf{H}^0 = \mathbf{X}_{de} \in \mathbb{R}^{\bar{s} \times d}$, the process in the l -th ACDecoder layer is

$$\mathbf{H}_1^l = \text{LN}(\mathbf{H}^{l-1} + \text{ACAttention}(\mathbf{H}^{l-1}, \mathbf{H}^{l-1}, \mathbf{H}^{l-1}, \hat{\mathbf{E}}^{ac})), \quad (18)$$

$$\mathbf{H}_2^l = \text{LN}(\mathbf{H}_1^l + \text{ACAttention}(\mathbf{H}_1^l, \mathbf{T}, \mathbf{T}, \hat{\mathbf{E}}^{ac})), \quad (19)$$

$$\mathbf{H}^l = \text{LN}(\mathbf{H}_2^l + \text{FFN}(\mathbf{H}_2^l)), \quad l = 1, \dots, L_{de}, \quad (20)$$

where $\text{LN}(\cdot)$ denotes LayerNorm [4] and $\mathbf{H}^l \in \mathbb{R}^{\bar{s} \times d}$ is the query representation after l layers.

3.3 Meta Degradation Pattern Memory

Established battery knowledge [2] suggests that battery degradation trajectories share a small set of patterns across batteries. We call these shared trajectory prototypes *meta degradation patterns*, as they compose diverse real-world trajectories. Inspired by memory networks [38, 46], we propose a meta degradation pattern memory

(MDPM) to store and retrieve such prototypes. MDPM maintains N_{mem} learnable memory slots $\Omega \in \mathbb{R}^{N_{\text{mem}} \times d}$, where each slot $\Omega_i \in \mathbb{R}^d$ stores one vector representation of a meta degradation pattern.

Pattern retrieval. Given decoder output $\mathbf{H}^{L_{de}} \in \mathbb{R}^{s \times d}$, we transform it into a memory query for retrieving relevant patterns by cosine similarity:

$$\mathbf{q}_{\text{mem}} = \text{FFN}(\text{Flatten}(\mathbf{H}^{L_{de}})) \in \mathbb{R}^d, \quad (21)$$

$$s_i = \frac{\mathbf{q}_{\text{mem}}^\top \Omega_i}{\|\mathbf{q}_{\text{mem}}\|_2 \|\Omega_i\|_2}, \quad i = 1, \dots, N_{\text{mem}}. \quad (22)$$

We select the top-2 memory slots with the largest similarity scores. Let $\mathcal{I}_2$ denote the corresponding index set. The relevant pattern embedding $\mathbf{h}_{\text{mem}}$ is retrieved as follows:

$$\alpha_i = \frac{\exp(s_i)}{\sum_{k \in \mathcal{I}_2} \exp(s_k)}, \quad i \in \mathcal{I}_2, \quad (23)$$

$$\mathbf{h}_{\text{mem}} = \sum_{i \in \mathcal{I}_2} \alpha_i \Omega_i \in \mathbb{R}^d. \quad (24)$$

Memory learning. During training, we encourage the retrieved pattern embedding $\mathbf{h}_{\text{mem}}$ to align with a full-life trajectory embedding $\mathbf{e}_{\text{trajectory}}$:

$$\mathcal{L}_{\text{align}} = \frac{1}{N} \sum_{i=1}^N \left(1 - \frac{\mathbf{h}_{\text{mem},i} \cdot \mathbf{e}_{\text{trajectory},i}}{\|\mathbf{h}_{\text{mem},i}\|_2 \|\mathbf{e}_{\text{trajectory},i}\|_2} \right), \quad (25)$$

where N is the batch size, $\mathbf{e}_{\text{trajectory},i} = \text{TrajectoryEncoder}(\mathbf{y}_i)$, and $\mathbf{h}_{\text{mem},i}$ is the retrieved pattern embedding for sample i .

To ensure $\mathbf{e}_{\text{trajectory}}$ preserves trajectory information, we reconstruct the trajectory with a decoder:

$$\bar{\mathbf{y}} = \text{TrajectoryDecoder}(\mathbf{e}_{\text{trajectory}}), \quad (26)$$

$$\mathcal{L}_{\text{recover}} = \frac{1}{N} \sum_{i=1}^N \frac{1}{O_i} \sum_{j=1}^{T_{\text{max}}} \text{mask}_{ij} (\mathbf{y}_{ij} - \bar{\mathbf{y}}_{ij})^2, \quad (27)$$

where $T_{\text{max}} = 5000$ is the maximum horizon, set to cover the longest degradation trajectories in the database, $\text{mask}_{ij} \in \{0, 1\}$ indicates whether the ground-truth SOH y_{ij} is available at cycle j for sample i and falls in the prediction region, and $O_i = \sum_{j=1}^{T_{\text{max}}} \text{mask}_{ij}$ is the number of observed SOH measurements. Both $\text{TrajectoryEncoder}(\cdot)$ and $\text{TrajectoryDecoder}(\cdot)$ are feed-forward networks with GELU.

Fusion and prediction. We incorporate the retrieved degradation pattern into the forecasting head via gated fusion:

$$\bar{\mathbf{H}} = \text{GELU}(\text{Flatten}(\mathbf{H}^{L_{de}} \mathbf{W}_3 + \mathbf{b}_3)) \mathbf{W}_4 + \mathbf{b}_4, \quad (28)$$

$$\boldsymbol{\beta} = \text{Sigmoid}(\text{FFN}([\bar{\mathbf{H}}; \mathbf{h}_{\text{mem}}])), \quad (29)$$

$$\hat{\mathbf{y}} = \text{Head}(\bar{\mathbf{H}} + \boldsymbol{\beta} \odot \mathbf{h}_{\text{mem}}), \quad (30)$$

where $\boldsymbol{\beta} \in \mathbb{R}^d$ is a feature-wise gate, $\odot$ denotes element-wise multiplication, and $\text{Head}(\cdot)$ is a linear projection that outputs the predicted degradation trajectory $\hat{\mathbf{y}}$.

Table 1: Statistics of four battery domains.

	Li-ion	CALB	Na-ion	Zn-ion
Number of batteries	963	27	31	95
Aging conditions	466	4	12	95
Charge and discharge protocols	398	2	12	1
Chemical systems	15	1	1	45
Operating temperatures	8	4	1	3
Cycle Average	863	866	185	410
Cycle Standard Deviation	717	496	53	399
Cycle Minimum	108	104	102	104
Cycle Maximum	4904	1411	308	1652

3.4 Training of BatteryMFormer

BatteryMFormer is trained with the following objective:

$$\min_{\theta} \mathcal{L}(\theta) = \mathcal{L}_{\text{pred}} + \lambda_1 \mathcal{L}_{\text{align}} + \lambda_2 \mathcal{L}_{\text{recover}}, \quad (31)$$

$$\mathcal{L}_{\text{pred}} = \frac{1}{N} \sum_{i=1}^N \frac{1}{O_i} \sum_{j=1}^{T_{\text{max}}} \text{mask}_{ij} (\mathbf{y}_{ij} - \hat{\mathbf{y}}_{ij})^2, \quad (32)$$

where λ_1 and λ_2 weight the alignment and recovery losses, respectively.

4 Experiments

4.1 Experimental Settings

Datasets. We evaluate our model and baselines on four battery domains from the largest public real-world battery lifetime database [36]. Dataset statistics are reported in Table 1.

- **Li-ion.** This domain contains lab-tested lithium-ion batteries (LIBs) aggregated from 13 subdatasets [1, 3, 7, 9, 11, 16, 17, 21, 22, 27, 29, 31, 33, 43, 45, 47, 50, 57]. Most batteries are commercial, covering diverse operating conditions and widely used LIB chemistries.
- **CALB.** This domain consists of large-format commercial LIBs tested in a production environment [36]. Compared with Li-ion, CALB reflects industrial development toward larger capacities and package structure.
- **Na-ion.** This domain includes commercial sodium-ion batteries evaluated under diverse charge and discharge protocols [36].
- **Zn-ion.** This domain contains zinc-ion batteries with varying electrolyte compositions and package structures, tested under different operating temperatures [36].

Metrics and dataset splits. In line with prior work [32, 41], we evaluate performance using mean absolute error (MAE) and mean absolute percentage error (MAPE), both computed on the original SOH values. We assess model generalizability under aging-condition-exclusive testing, where all test batteries come from aging conditions unseen during training and validation. For the Li-ion and Zn-ion domains, we generate three random splits while keeping the aging condition counts close to a 6:2:2 train/validation/test ratio. For CALB and Na-ion, where the number of aging conditions is limited, we use a leave-one-aging-condition-out protocol: one aging condition is held out for testing, and 25% of the remaining aging conditions are selected for validation while the rest are used for training. We report the mean and standard deviation over the resulting splits for each domain.

Table 2: Overall model performance on four battery domains. The top-three results are shaded. The best results are shown in bold and the second-best results are underlined. The improvement denotes the relative improvement of BatteryMFormer over the second-best model.

Model	Li-ion		CALB		Na-ion		Zn-ion	
	MAPE (%)	MAE (10^{-2})	MAPE (%)	MAE (10^{-2})	MAPE (%)	MAE (10^{-2})	MAPE (%)	MAE (10^{-2})
PatchTST	3.699 $\pm$ 0.269	3.385 $\pm$ 0.272	3.096 $\pm$ 2.703	2.906 $\pm$ 2.468	1.783 $\pm$ 0.557	1.484 $\pm$ 0.461	5.969 $\pm$ 0.630	5.773 $\pm$ 0.528
DLinear	4.155 $\pm$ 0.336	3.802 $\pm$ 0.265	17.968 $\pm$ 23.386	16.429 $\pm$ 21.273	1.607 $\pm$ 0.560	1.331 $\pm$ 0.457	7.288 $\pm$ 0.821	7.029 $\pm$ 0.649
iTransformer	2.986 $\pm$ 0.182	2.727 $\pm$ 0.190	3.203 $\pm$ 2.754	3.004 $\pm$ 2.518	1.884 $\pm$ 0.737	1.564 $\pm$ 0.605	5.938 $\pm$ 0.673	5.743 $\pm$ 0.442
TimesFM	8.668 $\pm$ 0.311	7.680 $\pm$ 0.248	4.220 $\pm$ 1.329	3.950 $\pm$ 1.264	4.100 $\pm$ 0.731	3.378 $\pm$ 0.620	14.819 $\pm$ 0.778	13.615 $\pm$ 0.547
ConvTimeNet	3.472 $\pm$ 0.236	3.157 $\pm$ 0.169	3.255 $\pm$ 2.790	3.044 $\pm$ 2.555	1.491 $\pm$ 0.653	1.233 $\pm$ 0.536	7.810 $\pm$ 0.563	7.628 $\pm$ 0.829
TimeMixer++	2.690 $\pm$ 0.135	2.446 $\pm$ 0.131	3.281 $\pm$ 2.268	3.075 $\pm$ 2.064	1.303 $\pm$ 0.461	1.082 $\pm$ 0.385	6.749 $\pm$ 0.953	6.593 $\pm$ 0.908
PatchMLP	2.819 $\pm$ 0.023	2.561 $\pm$ 0.046	23.502 $\pm$ 21.780	21.888 $\pm$ 19.887	1.350 $\pm$ 0.348	1.120 $\pm$ 0.289	6.140 $\pm$ 1.254	5.835 $\pm$ 1.069
CPMLP	2.615 $\pm$ 0.073	2.380 $\pm$ 0.067	3.371 $\pm$ 2.842	3.161 $\pm$ 2.597	1.267 $\pm$ 0.484	1.050 $\pm$ 0.405	6.841 $\pm$ 0.554	6.733 $\pm$ 0.623
CPTransformer	2.557 $\pm$ 0.121	2.325 $\pm$ 0.118	3.354 $\pm$ 2.784	3.141 $\pm$ 2.549	1.511 $\pm$ 0.589	1.254 $\pm$ 0.485	5.811 $\pm$ 0.589	5.649 $\pm$ 0.512
TimeBridge	3.037 $\pm$ 0.446	2.776 $\pm$ 0.438	2.287 $\pm$ 1.296	2.136 $\pm$ 1.186	1.217 $\pm$ 0.517	1.008 $\pm$ 0.436	5.563 $\pm$ 0.403	5.408 $\pm$ 0.219
IC2ML	2.528 $\pm$ 0.220	2.284 $\pm$ 0.206	1.955 $\pm$ 1.212	1.892 $\pm$ 1.156	1.348 $\pm$ 0.976	1.114 $\pm$ 0.801	5.460 $\pm$ 0.283	5.355 $\pm$ 0.197
BatteryMFormer	2.248 $\pm$ 0.037	2.034 $\pm$ 0.041	1.789 $\pm$ 0.282	1.687 $\pm$ 0.276	1.002 $\pm$ 0.446	0.830 $\pm$ 0.371	4.970 $\pm$ 0.654	4.721 $\pm$ 0.618
Improvement	11.07%	10.94%	8.49%	10.83%	17.66%	17.65%	8.97%	11.83%

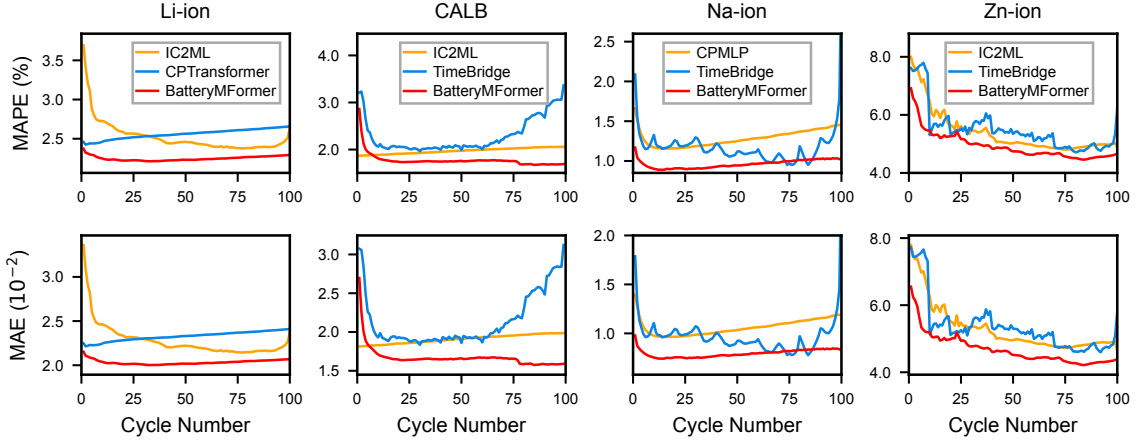


Figure 4: Performance of the top-three models as the number of usable early cycles increases.

Baselines. We compare against state-of-the-art methods in two groups. (1) Battery-specific BDTF models: IC2ML [14], CPTransformer [36], and CPMLP [36]. (2) Generic time-series forecasting models: Transformer-based methods (TimeMixer++ [44], TimeBridge [24], iTransformer [26], TimesFM [8], and PatchTST [30]), multi-layer perceptron (MLP) methods (PatchMLP [39] and DLinear [49]), and convolutional neural network (CNN)-based methods (ConvTimeNet [6]). Following the BatteryLife benchmark protocol [36], all baselines take voltage, current, and capacity sequences as input and predict the future degradation trajectory, except IC2ML and TimesFM. Following the original designs, IC2ML uses only the charging capacity-increment sequence as input, and TimesFM extrapolates future SOH values from the historical SOH sequence.

Implementation details. Following prior work [36], we re-sample each cycle to a unified length of $L = 300$. All models are implemented in PyTorch and trained for up to 300 epochs with early stopping (patience 30) based on validation performance. For

each trainable model and domain, we evaluate at least 10 hyperparameter configurations and report the one with the best validation performance. All experiments are conducted on NVIDIA RTX 3090 GPUs. Additional implementation and preprocessing details are provided in Appendix C and Appendix D, respectively.

4.2 Overall Performance

4.2.1 Main results. Table 2 compares BatteryMFormer with state-of-the-art baselines across four battery domains. BatteryMFormer achieves the best performance on all domains and metrics despite substantial differences in battery chemistry, data scale, and degradation characteristics. Compared with the second-best model in each domain, BatteryMFormer reduces MAPE by 11.07%, 8.49%, 17.66%, and 8.97%, and reduces MAE by 10.94%, 10.83%, 17.65%, and 11.83% on Li-ion, CALB, Na-ion, and Zn-ion, respectively. These consistent improvements demonstrate the effectiveness of BatteryMFormer for early BDTF across domains.

Table 3: Ablation study of BatteryMFormer on four battery domains.

Model	Li-ion		CALB		Na-ion		Zn-ion	
	MAPE (%)	MAE (10^{-2})	MAPE (%)	MAE (10^{-2})	MAPE (%)	MAE (10^{-2})	MAPE (%)	MAE (10^{-2})
BatteryMFormer	2.248 ± 0.037	2.034 ± 0.041	1.789 ± 0.282	1.687 ± 0.276	1.002 ± 0.446	0.830 ± 0.371	4.970 ± 0.654	4.721 ± 0.618
w/o SOCVIEW	2.726 ± 0.112	2.476 ± 0.113	2.010 ± 0.681	1.898 ± 0.615	1.393 ± 0.399	1.156 ± 0.331	6.647 ± 1.309	6.343 ± 1.193
w/o MDPM	2.497 ± 0.172	2.268 ± 0.168	2.158 ± 0.878	2.031 ± 0.788	1.252 ± 0.459	1.038 ± 0.383	6.133 ± 1.705	5.924 ± 1.597
w/o ACDecoder	2.560 ± 0.042	2.328 ± 0.048	2.449 ± 1.334	2.301 ± 1.209	1.243 ± 0.422	1.029 ± 0.351	7.722 ± 0.384	7.516 ± 0.601
w/o ACAttention	2.488 ± 0.045	2.261 ± 0.035	2.058 ± 0.743	1.941 ± 0.670	1.229 ± 0.401	1.018 ± 0.332	7.602 ± 1.009	7.446 ± 1.181
w/o ACQuery	2.540 ± 0.060	2.306 ± 0.040	2.263 ± 1.007	2.129 ± 0.905	1.226 ± 0.376	1.016 ± 0.314	6.324 ± 1.369	6.095 ± 1.299
w/o LLM	2.270 ± 0.167	2.056 ± 0.157	3.151 ± 2.938	2.948 ± 2.693	1.011 ± 0.472	0.839 ± 0.395	5.614 ± 0.940	5.369 ± 0.896
CPTransformer-SI	2.629 ± 0.173	2.391 ± 0.170	3.162 ± 2.641	2.957 ± 2.419	1.569 ± 0.517	1.302 ± 0.430	5.581 ± 0.436	5.358 ± 0.366

Notably, the strongest baseline varies across domains: IC2ML performs best among baselines on Li-ion, CALB, and Zn-ion, while TimeBridge achieves the best baseline performance on Na-ion. This suggests that the predictive patterns underlying battery degradation are domain-dependent, and different architectural inductive biases match these patterns to different extents. This heterogeneity regarding the underlying effective patterns can also lead to pronounced performance instability. For example, DLinear performs competitively on Na-ion but incurs much larger errors on CALB, and TimesFM shows relatively high errors on Li-ion, Na-ion and Zn-ion. In contrast, BatteryMFormer consistently achieves the best performance across domains, indicating that it can capture a broader spectrum of degradation patterns in different battery datasets.

4.2.2 Comparison under different numbers of usable cycles. We further evaluate model performance by varying the number of usable early cycles. Figure 4 reports MAE and MAPE for the top-performing models on each domain. BatteryMFormer achieves consistent improvements across a broad range of early-forecasting settings across four domains. These results demonstrate the effectiveness of BatteryMFormer under different amounts of early degradation information.

We also observe that prediction errors can increase on Li-ion and Na-ion when $S > 25$. This reflects an open challenge in long-sequence time-series modeling: longer inputs do not guarantee improvements, and may instead introduce redundancy and optimization difficulty. In our setting, each cycle contains 300 points, so $S > 25$ already yields more than 7,500 input points. Since adjacent battery cycles often change only marginally, longer inputs may dilute informative degradation signatures with redundant measurements. Similar error increases are also observed in other top-performing baselines and broader time-series forecasting studies when modeling long input sequences [30, 39]. Despite this effect, BatteryMFormer generally outperforms the baselines, underscoring the advantage of the proposed multi-level learning strategy in handling long operational voltage and current time series for early BDTF.

4.3 Ablation Study

We conduct an ablation study of BatteryMFormer to evaluate the effectiveness of its key components, with the results summarized in Table 3. "w/o SOCVIEW" removes the SOC view from the dual-view

encoder, and "w/o MDPM" removes the meta degradation pattern memory. For the aging-condition-aware decoder, "w/o ACQuery" removes aging-condition information from generic queries, "w/o ACAttention" replaces aging-condition-aware attention with standard attention, and "w/o ACDecoder" removes both mechanisms. "w/o LLM" replaces the language-based embedder with factor-wise lookup embeddings followed by projection. Since variable-length protocols, such as multi-stage charge/discharge protocols, cannot be trivially encoded by fixed lookup embeddings due to out-of-vocabulary issues, this variant only uses positive electrode, negative electrode, operating temperature, package structure, and manufacturer as lookup-embedding factors. "CPTransformer-SI" provides CPTransformer with the same input information as BatteryMFormer, including voltage, current, capacity, SOC, aging-condition information, and cycle-level descriptors.

Table 3 shows that the three major components of BatteryMFormer all contribute to performance. Removing the SOC view, MDPM, or ACDecoder consistently degrades results across all domains, indicating that learning SOC-localized patterns, trajectory-level prototypes, and aging-condition-informed representations is useful for early BDTF. Their contributions are domain-dependent: the SOC view brings particularly clear gains on Li-ion and Na-ion, while ACDecoder has more pronounced effects on CALB and Zn-ion. This suggests that the effective predictive patterns vary across battery domains, and that the components of BatteryMFormer capture complementary aspects of these patterns. CPTransformer-SI remains clearly worse than BatteryMFormer and performs close to CPTransformer in most domains, indicating that simply providing the same input information yields limited benefits. The advantage of BatteryMFormer therefore comes primarily from its multi-level learning architecture rather than additional input variables alone. Finer-grained ACDecoder ablations further show that both ACQuery and ACAttention are important for aging-condition-aware pattern mining. Finally, replacing the LLM embedder with lookup embeddings consistently degrades performance, confirming that semantic aging-condition representations provide useful information beyond factor-wise embeddings. While the mean difference is small on Li-ion and Na-ion, the larger standard deviations of w/o LLM suggest that the LLM embedder improves performance stability; on CALB and Zn-ion, it improves both accuracy and robustness. Collectively, these results validate the effectiveness of the proposed multi-level learning strategy.

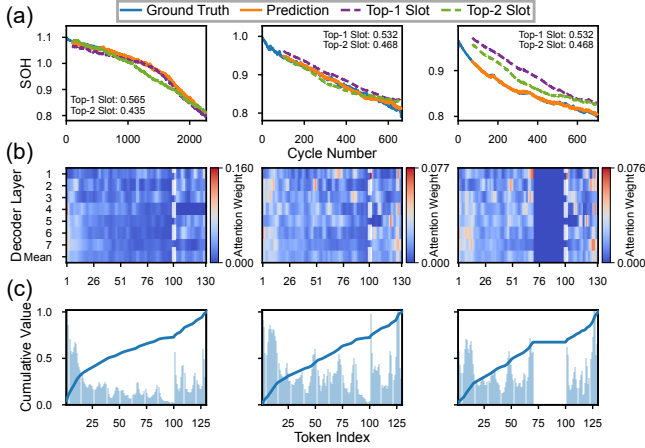


Figure 5: Case study on three representative test batteries with superlinear, linear, and sublinear degradation. (a) Ground-truth vs. predicted SOH trajectories with the top two retrieved MDPM prototypes (weights shown). (b) Attention weights of ACDecoder cross-attention over temporal-view and SOC-view tokens; dashed line marks the boundary. (c) Token-wise attention weights (bars) and cumulative weight (lines) over token indices.

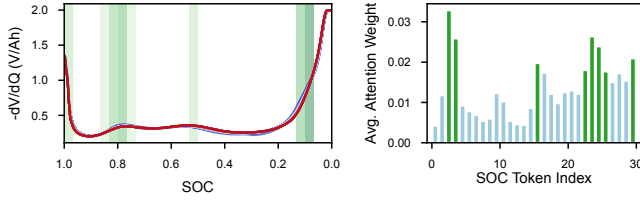


Figure 6: Differential voltage analysis and average attention weights of SOC-view tokens for a representative test battery. Highlighted regions denote the SOC intervals corresponding to the top-25% SOC-view tokens ranked by attention weight, with darker green indicating larger attention weights.

4.4 Case Study

To interpret how BatteryMFormer leverages the multi-level structure of early BDTF, we examine three representative test batteries exhibiting superlinear, linear, and sublinear degradation (Figure 5). We analyze (i) the top-two meta degradation patterns retrieved from MDPM, visualized by decoding the corresponding memory embeddings with the trajectory decoder, and (ii) the ACDecoder cross-attention weights over temporal-view and SOC-view tokens.

Figure 5a shows that MDPM retrieves prototypes that are consistent with the batteries’ long-horizon degradation patterns. For the superlinear battery, the retrieved prototypes include two trajectory prototypes showing accelerated degradation with knee points, providing informative priors for long-range extrapolation. For the linear battery, the retrieved prototypes are approximately linear and largely agree with the observed trend. For the sublinear battery, MDPM retrieves prototypes exhibiting a slowdown in degradation

Table 4: Results of data-efficient learning with 50% of the training batteries retained. Top-3 and Top-2 denote the third-best and second-best baselines selected from the overall comparison.

Dataset	Metrics	Top-3	Top-2	BatteryMFormer
Li-ion	MAPE (%)	2.958 $\pm$ 0.134	2.802 $\pm$ 0.320	2.453 $\pm$ 0.096
	MAE (10^{-2})	2.696 $\pm$ 0.104	2.530 $\pm$ 0.297	2.222 $\pm$ 0.077
CALB	MAPE (%)	2.426 $\pm$ 1.072	2.237 $\pm$ 1.573	2.174 $\pm$ 0.905
	MAE (10^{-2})	2.272 $\pm$ 0.986	2.162 $\pm$ 1.511	2.049 $\pm$ 0.832
Na-ion	MAPE (%)	1.497 $\pm$ 0.538	1.234 $\pm$ 0.360	1.046 $\pm$ 0.455
	MAE (10^{-2})	1.245 $\pm$ 0.448	1.021 $\pm$ 0.296	0.868 $\pm$ 0.381
Zn-ion	MAPE (%)	7.742 $\pm$ 1.115	7.545 $\pm$ 0.699	6.210 $\pm$ 0.334
	MAE (10^{-2})	7.426 $\pm$ 0.873	7.402 $\pm$ 1.040	6.066 $\pm$ 0.204

even though the input covers only the early, faster-decay stage, indicating that the MDPM stores diverse global trajectory shapes and can retrieve related trajectory prototypes to help BDTF for batteries subjected to aging conditions not covered by the training data.

Figure 5b–c further indicates that ACDecoder integrates both views. Most attention mass is assigned to temporal-view tokens, while SOC-view tokens receive a non-trivial share. Moreover, attention over SOC-view tokens is highly concentrated on a small subset, suggesting that degradation-relevant operational signatures are localized to specific SOC intervals and that BatteryMFormer can prioritize these regions through attention. Overall, this case study illustrates how MDPM supplies pattern-level priors and how the dual-view encoder with ACDecoder selectively aggregates temporal and SOC-localized cues to improve early BDTF.

We further interpret the SOC-view attention through differential voltage analysis (DVA) on a test battery (Figure 6). The top-25% SOC tokens ranked by attention weights are concentrated around the major DVA peaks and their shoulder regions, which are known to be sensitive to degradation-induced changes such as peak shifts, broadening, and shape distortion caused by lithium inventory loss, loss of active material, or polarization growth [5, 37]. This result suggests that the SOC view guides BatteryMFormer to selectively attend to particular SOC intervals. The alignment with DVA features further indicates that the learned SOC-localized patterns can reflect electrochemical signatures associated with battery aging mechanisms.

4.5 Data-Efficient Learning

Collecting full-life degradation trajectories is costly and can take months to years, making data-efficient BDTF an important practical requirement. To evaluate model robustness under limited lifetime data, we retain only 50% of the training batteries in each domain while keeping the validation and test parts unchanged. Table 4 reports the results of BatteryMFormer and the top-performing baselines under this reduced-data setting.

BatteryMFormer achieves the best performance across all four domains with only 50% of the training data. Compared with the strongest baseline, it reduces MAPE by 12.45%, 2.81%, 15.23%, and 17.69%, and reduces MAE by 12.17%, 5.22%, 14.98%, and 18.04% on Li-ion, CALB, Na-ion, and Zn-ion, respectively. The improvements

are particularly pronounced on Na-ion and Zn-ion, where training data are limited and aging conditions are diverse. These results indicate that the proposed multi-level learning strategy can still extract informative degradation patterns from limited lifetime data, thereby improving data efficiency in early BDTF.

5 Related Work

We review existing approaches on battery degradation trajectory forecasting (BDTF) from two perspectives: feature engineering and representation learning.

Feature-engineering-based methods. These methods focus on extracting degradation-relevant descriptors from operational measurements (e.g., voltage, current, capacity, relaxation signals) and then fit data-driven predictors for future capacity/SOH trajectories [19, 28, 41]. A common practice is to extract features from regions where aging signatures are pronounced. For instance, [19] constructs descriptors from the late-charge capacity sequence and the post-charge relaxation voltage region and then uses an LSTM as the forecaster; [41] designs features tailored to a 9-step charging protocol and applies a feed-forward network for trajectory prediction. While effective on curated settings, these handcrafted descriptors are often protocol- or dataset-specific (e.g., tied to particular voltage windows or multi-step procedures) and may not be available or predictive across diverse aging conditions [27, 36].

Representation-learning-based methods. In contrast, methods in this research line learn forecasting-relevant representations directly from raw or minimally processed measurements using neural networks [14, 15, 22, 23, 25, 36, 37]. An intuitive strategy treats BDTF as generic time-series forecasting by extrapolating future SOH from historical SOH records using architectures such as LSTM or Transformer variants [22, 23, 34, 37]. However, SOH-only inputs can be weakly informative in early cycles, where early trajectories appear similar yet diverge substantially in the long horizon. Recent work therefore exploits fine-grained voltage-current profiles. The BatteryLife benchmark [36] shows that directly applying generic forecasters for modeling voltage and current time series can be suboptimal, and introduces CyclePatch to model intra-cycle dynamics and inter-cycle evolution more effectively; IC2ML [14] further improves this paradigm by injecting auxiliary supervision to enhance learned representations. Our BatteryMFormer belongs to this representation-learning family, but advances beyond existing approaches by explicitly integrating multi-level inductive biases for early BDTF.

6 Limitations and Ethical Considerations

Limitations. First, using more early cycles yields long and redundant inputs; for example, more than 25 cycles already corresponds to over 7,500 input points. This can compromise model performance on ultra-long inputs. Second, we evaluate on regular laboratory/production tests that are critical for battery optimization and production, whereas field data (e.g., EV logs) are often irregular and noisier due to varying usage and sensor noise. Applying BatteryMFormer to field conditions may require modified representations and preprocessing (e.g., handling inaccurate and irregular data records [52]).

Ethical considerations. This work utilizes publicly available battery lifetime datasets and contains no human-subject or personally identifiable data. In field deployment, forecast errors can induce suboptimal decisions (e.g., premature retirement or delayed maintenance), so models should be validated against the target operating distribution before high-stakes deployment.

7 Conclusion and Future Work

This paper highlights the value of explicitly modeling the multi-level structure in early battery degradation trajectory forecasting, spanning trajectory patterns, aging conditions, and battery-specific dynamics. We propose BatteryMFormer and demonstrate that our model delivers consistent improvements over state-of-the-art baselines across four battery domains. Ablation and case studies further confirm that each component contributes meaningfully to these gains. BatteryMFormer also performs better in data-efficient settings under reduced training data. Future endeavors will focus on improving the model’s ability to model long operational time series and adapting the framework to irregular, noisy field data.

8 GenAI Disclosure

Generative AI tools were used to assist with language editing (e.g., improving clarity, grammar, and conciseness) of author-written text and to support code development (e.g., drafting or refactoring implementation snippets). These tools were not used to generate the experimental results reported in this work. All AI-assisted edits to the manuscript and code were reviewed and validated by the authors, who take full responsibility for the correctness, originality, and integrity of the work.

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A End-of-life Definition

We define end-of-life t_{eol} as the first cycle at which SOH falls below the threshold τ . We use $\tau = 80\%$ for Li-ion, Na-ion, and Zn-ion. For CALB, since many batteries are not degraded to 80% SOH within the measured data, we use $\tau = 90\%$ following BatteryLife [36].

B Details of SOC Calculation

Each battery provides a protocol SOC interval $[\text{SOC}_{\text{start}}^{\text{ch}}, \text{SOC}_{\text{end}}^{\text{ch}}]$, where charging starts at $\text{SOC}_{\text{start}}^{\text{ch}}$ and ends at $\text{SOC}_{\text{end}}^{\text{ch}}$; the subsequent discharge returns to $\text{SOC}_{\text{start}}^{\text{ch}}$ before the next cycle.

Within each cycle i , we assume SOC varies linearly with the within-segment charge/discharge capacity change. Let $Q_{i,k}$ denote the capacity at point k in cycle i . For charging, with segment end-points $Q_{i,\text{start}}^{\text{ch}}$ and $Q_{i,\text{end}}^{\text{ch}}$, we compute

$$\text{SOC}_{i,k} = \text{SOC}_{\text{start}}^{\text{ch}} + \frac{Q_{i,k} - Q_{i,\text{start}}^{\text{ch}}}{Q_{i,\text{end}}^{\text{ch}} - Q_{i,\text{start}}^{\text{ch}}} \left(\text{SOC}_{\text{end}}^{\text{ch}} - \text{SOC}_{\text{start}}^{\text{ch}} \right). \quad (33)$$

For discharging, we use the same linear mapping but reverse the SOC direction from $\text{SOC}_{\text{end}}^{\text{ch}}$ to $\text{SOC}_{\text{start}}^{\text{ch}}$:

$$\text{SOC}_{i,k} = \text{SOC}_{\text{end}}^{\text{ch}} + \frac{Q_{i,k} - Q_{i,\text{start}}^{\text{dis}}}{Q_{i,\text{end}}^{\text{dis}} - Q_{i,\text{start}}^{\text{dis}}} \left(\text{SOC}_{\text{start}}^{\text{ch}} - \text{SOC}_{\text{end}}^{\text{ch}} \right). \quad (34)$$

SOC-aligned resampling. We re-parameterize each segment by SOC and resample voltage, current, and capacity on a uniform SOC grid. Concretely, for charging we interpolate each variable at $L/2$ equally spaced SOC values in $[\text{SOC}_{\text{start}}^{\text{ch}}, \text{SOC}_{\text{end}}^{\text{ch}}]$; for discharging we use $L/2$ equally spaced values in the reverse direction. We then concatenate the resampled charging and discharging sequences to form a length- L per-cycle input, where the k -th point corresponds to a fixed SOC level (within the recorded interval), yielding SOC-aligned inputs for BatteryMFormer.

C Further Implementation Details

This appendix provides additional implementation details to facilitate reproducibility. We first describe the unified input processing pipeline used for all baselines that leverage cycling data, and then explain how we adapt generic time-series forecasters and battery-specific baselines to the early BDTF setting.

C.1 Input and Target Construction

Input processing. For each battery, we process the raw cycling record on a per-cycle basis. For cycle i , we resample the charging and discharging segments to $L/2$ uniformly spaced points (per segment) and concatenate them in the order *charge* $\rightarrow$ *discharge*, yielding length- L sequences for voltage, current, and capacity. Let $\mathbf{v}_i \in \mathbb{R}^L$, $\mathbf{I}_i \in \mathbb{R}^L$, and $\mathbf{c}_i \in \mathbb{R}^L$ denote the resampled voltage, current, and capacity, respectively. Following prior work [36], we normalize current to C-rate by dividing by the (per-battery) nominal capacity:

$$\mathbf{I}_i \leftarrow \mathbf{I}_i / Q_{\text{nominal}}, \quad (35)$$

where Q_{nominal} is provided by the dataset. Voltage and capacity are kept in their original scales. We then form the per-cycle input as

$$\tilde{\mathbf{X}}_i = [\mathbf{v}_i; \mathbf{I}_i; \mathbf{c}_i] \in \mathbb{R}^{3 \times L}. \quad (36)$$

For BatteryMFormer, we additionally compute the SOC variable as described in Appendix B. All models are trained and evaluated under the early BDTF protocol in Section 2.3: we use at most the first 100 cycles as input, and if fewer cycles are available, we pad the missing cycles with all-zero sequences. Specifically, for a setting with $S \in \{1, \dots, 100\}$ usable early cycles, we build $\tilde{\mathbf{X}}_{1:100}$ by placing the available $\{\tilde{\mathbf{X}}_i\}_{i=1}^S$ in the first S cycles and zero-padding the remaining cycles. We also provide a cycle-level validity mask

$$\mathbf{m}^{\text{cyc}} \in \{0, 1\}^{100}, \quad (37)$$

where $m_i^{\text{cyc}} = 1$ if and only if cycle i exists (i.e., $i \leq S$) and 0 otherwise; models that support attention masking use $\mathbf{m}^{\text{cyc}}$ to ignore padded cycles.

Target normalization and padding. Each battery trajectory is padded to a maximum horizon of 5000 cycles, which covers the longest trajectories in the database. To account for domain-specific EOL thresholds, we normalize SOH as

$$\tilde{y}_j = \frac{y_j - \tau}{1 - \tau}, \quad (38)$$

where y_j is the ground-truth SOH at cycle j and τ is the EOL threshold defined in Appendix A. During training, Equation (32) is applied in the normalized SOH space using $\tilde{y}$ and $\hat{y}$. For evaluation, predictions are transformed back to the original SOH scale before computing MAE and MAPE.

C.2 Baseline Implementation

We next describe how we adapt baselines for battery degradation trajectory forecasting.

Generic time-series forecasting models. We reshape $\tilde{X}_{1:100} \in \mathbb{R}^{100 \times 3 \times L}$ as the time series inputs of the generic time-series forecasting models:

$$\mathbf{X}_{\text{flat}} = \text{Reshape}(\tilde{X}_{1:100}) \in \mathbb{R}^{(100 \cdot L) \times 3}. \quad (39)$$

The core architecture of the backbone encoder $f(\cdot)$ is kept unchanged from the original implementation, and we replace the original forecasting head with a trajectory prediction head that outputs a length-5000 SOH sequence:

$$\mathbf{H} = f(\mathbf{X}_{\text{flat}}), \quad (40)$$

$$\hat{\mathbf{y}} = \text{Head}(\mathbf{H}) \in \mathbb{R}^{5000}. \quad (41)$$

Here, $\text{Head}(\cdot)$ first flattens $\mathbf{H}$ if necessary and then applies a linear projection to forecast 5000 SOH points.

TimesFM is a pre-trained time-series foundation model that performs zero-shot forecasting without task-specific fine-tuning. Since TimesFM only supports univariate time-series inputs, we feed the historical SOH sequence directly into the model. Due to its patch-based architecture with a fixed patch length of 32, the input length must be a multiple of 32. If the length of the historical SOH sequence is not a multiple of 32, we pad the sequence to the next multiple of 32 by repeating the last observed SOH value. TimesFM then autoregressively generates the future SOH sequence up to the specified prediction horizon.

Battery-specific models. For CPTransformer and CPMLP, we follow the official BatteryLife implementations and only adjust the output head to produce the length-5000 trajectory. For IC2ML [14], the original method constructs the input from the charging capacity-increment sequence within a fixed 3.6–3.8 V voltage window. Because this fixed voltage window is not available or comparable for all batteries and protocols in our datasets, we adapt IC2ML by computing the capacity-increment sequence over each battery’s observed charging voltage range and use it as the model input. All other components follow the original paper and the official repository.

Training objective. For supervised baselines, we use the same masked-MSE prediction loss as BatteryMFormer (Equation 32). For IC2ML, we additionally include its original multi-task auxiliary losses. TimesFM is evaluated zero-shot without task-specific fine-tuning.

Hyperparameter search. For BatteryMFormer, we perform per-fold hyperparameter search using Bayesian optimization, running multiple trials per fold and selecting the configuration with the lowest validation MAPE. The search space includes learning rate in $[2 \times 10^{-5}, 2 \times 10^{-4}]$, batch size in $\{64, 128\}$, dropout rate in $[0.05, 0.5]$, embedding dimension $d \in \{64, 128, 256\}$, feed-forward

dimensions $d_{\text{ff}} \in \{32, 64, 128\}$ and $d_{\text{ffs}} \in \{32, 64, 128, 256\}$, key dimension in $\{512, 768\}$, memory dimension in $\{128, 512\}$, $L_{\text{intra}} \in \{2, 4\}$, $L_{\text{de}} \in \{2, 4, 6, 8\}$, number of queries in $\{4, 8, 10, 12, 20, 50\}$, $N_{\text{mem}} \in \{64, 96\}$, and patch-encoder kernel size in $\{10, 16, 20, 30\}$. For fair comparison, all models except the zero-shot TimesFM are tuned on the same training/validation splits with at least 10 configurations per domain, and the configuration with the lowest validation MAPE is selected for test evaluation.

D Further Details of Data Preprocessing

This section describes the preprocessing pipeline applied to all datasets. Raw operational data may contain non-standard segments, such as reference performance tests (RPTs), formation cycles, and equipment faults. These segments can introduce abrupt SOH deviations that are unrelated to the predominant cycling protocol and may impair model training if treated as regular degradation signals. We therefore smooth evident measurement artifacts and non-standard testing segments, while preserving sustained degradation trends.

SOH computation and battery filtering. We compute SOH as defined in Section 2.2. To suppress isolated measurement artifacts, single-cycle SOH drops exceeding 3% of the previous cycle’s SOH are clipped to the previous-cycle value. This operation is only applied to isolated spike-like drops and does not modify sustained degradation trends. Batteries whose SOH has not degraded below $\tau + 2.5\%$ are excluded due to insufficient degradation information, where $\tau = 90\%$ for CALB [36] and $\tau = 80\%$ for all other datasets (Appendix A). For batteries that have degraded to $\tau + 2.5\%$ but have not yet reached τ , we estimate the EOL point by linearly extrapolating the last 20 measured cycles to τ . Measured SOH values are used whenever available; the extrapolated segment is only used to complete the trajectory up to the estimated EOL point for consistent target construction.

SOH trajectory smoothing. We identify artifact onsets at the region level using dataset-specific information. For datasets with RPT timestamps (ISU-ILCC [21]), each RPT start time is mapped to a cycle index, and the last normal cycle before the RPT is used as the onset. For datasets with cycle timestamps but no RPT annotations (HNEI [9], RWTH [22], Tongji [57], MICH_EXP [29]), cycle k is flagged as an onset when $(t_k - t_{k-1}) > \gamma_{\text{gap}}$, where γ_{gap} is fixed for each dataset before model training. For the remaining datasets, cycle k is flagged when $\delta_k > \gamma^+$ or $\delta_k < \gamma^-$, where $\delta_k = (\text{SOH}_k - \text{SOH}_{k-1})/\text{SOH}_{k-1}$, and γ^+ and γ^- are the 99th and 1st percentiles of the empirical $\{\delta_k\}$ distribution computed on the training split.

After an artifact onset k_s is identified, we locate the recovery point k_e by scanning forward and selecting the earliest cycle whose SOH returns within a tolerance ϵ of the pre-artifact level SOH_{k_s-1} and remains within this tolerance for the next W consecutive cycles. The affected region $[k_s, k_e]$ is then smoothed using PCHIP (Piecewise Cubic Hermite Interpolating Polynomial) interpolation over cycle indices, with M normal cycles immediately before and after the region used as anchor points. The same preprocessing pipeline is applied consistently across training, validation, and test splits.